Organisms and Populations

Introduction and Organism and Its Environment

1. Refer to the given analogy and select the option that fills the blanks correctly in it.
Eurythermal: Man : : Stenothermal : _____
1) Frog
2) Goat
3) Tiger
4) Cow

2. Temporary movement of animals from a stressful habitat to a hospitable area is called
1) Hibernation
2) Suspension
3) Migration
4) Navigation

3. Why does cactus have so many thorns?
1) For photosynthesis and storage of water
2) For protection from predators and conservation of water
3) For photosynthesis and protection
4) None of the above

4. Major biomes of India include
1) Tropical rain forest
2) Deciduous forest
3) Desert
4) All of the above

5. In habitats such as thermal springs and -deep-sea hydrothermal vents the average temperatures exceeds
1) 20 degree Celsius
2) 40 degree Celsius
3) 100 degree Celsius
4) 5 degree Celsius

6. Identify the mismatched pair.
1) Hibernation- Bear
2) Migration- Siberian crane
3) Aestivation- Snails
4) Diapause-Penguin

7. Thick layer of fat called blubber, found under the skin is characteristic feature of
1) Camel
2) Kangaroo rat
3) Seal
4) Siberian crane

8. When was Ramdeo Misra born?
1) 13th April, 1980
2) 26th August, 1908
3) 5th December, 1914
4) 23rd June, 1951

9. Ramdeo Misra is known as _____ in India.
1) Father of Ecology
2) Father of Botany
3) Father of Genetics
4) Father of Zoology

10. Extreme and harsh habitats inhabited by microbes include
I. Scorching desert of Rajasthan
II. Rain soaked forests of Meghalaya
III. Deep ocean trenches
IV. Permafrost Polar regions
V. Thermal springs
VI. High mountain tops
1) I, III and V
2) II, III, IV and V
3) II, III, IV and V
4) I, II, III, IV, V and VI

11. Mango trees are not found in
1) Canada
2) Germany
3) Temperate countries
4) All of the above

12. Light is an important abiotic factor because
1) Plants need sunlight for photosynthesis.
2) Plants are dependent on sunlight to meet their photoperiodic requirements for flowering.
3) Many animals use the diurnal and seasonal variations in light intensity and duration (photoperiod) as cues for timing their foraging, reproductive and migratory activities.
4) All of the above

13. Which of the following animals are homeothermic?
1) Snake
2) Sparrow
3) Tortoise
4) Salmon

14. Which of the following is a behavioural adaptation?
1) Animals inhabiting cold areas have thick fur coat.
2) Camel excretes concentrated urine to conserve water.
3) Desert lizard basks in Sun when the body temperature drops below comfort zone but moves into shade when ambient temperature starts increasing.
4) In desert plant Cactus, leaves are reduced to spine for protection and to conserve water.

15. Which animal migrates to temporarily escape from harsh environmental conditions?
1) Polar bear
2) Siberian crane
3) Camel
4) Kagaroo rat

16. Hibernation refers to
1) Winter sleep
2) Summer sleep
3) Mass movement from unfavourable place to favourable place
4) Both (1) and (2)

17. Which animal does not show migration?
1) Siberian crane
2) Arctic tern
3) Great white pelican
4) Pigeon

18. Which is an example of acclimatisation?
1) A hognose snake playing dead on sensing danger.
2) Arctic tern travels from Arctic region to Antarctic region to escape harsh winters.
3) Increase in number of RBCs in blood of a person who has gone to visit a hill station from plains during his vacations.
4) Aquatic mammals like seal and walrus have thick layer of fat called blubber under their skin.

19. If you visit a place like Ladakh in vacations, what changes will happen in your body?
I. Increase in RBC count
II. Decrease in RBC count
III. Decrease in binding affinity of haemoglobin
IV. Increase in binding affinity of haemoglobin
V. Increase in breathing rate
VI. Decrease in breathing rate
1) I, II and V
2) I, III and V
3) II, IV and VI
4) II, III and IV

20. Mammals of colder climates have shorter ears and limbs to
1) Maximize heat loss
2) Increase surface area so that excess heat may be easily dissipated
3) Minimise heat loss
4) None of the above

21. Why night blooming flowers are generally white?
1) Insects and other pollinators could easily spot plants with white flowers against the dark background.
2) Plants with white flowers are mostly parasites so they do not need sunlight for photosynthesis, hence such plants prefer to bloom at night.
3) White flowers attract diurnal insects and other small animals during night time.
4) None of the above

22. Ecology is a subject which studies
1) Internal structure of plants and animals
2) Evolutionary history of different related taxa
3) Interaction among organisms and between organism and its physical environment
4) Various celestial bodies in universe

23. Mean annual precipitation has the highest range in which of the following biomes?
1) Desert
2) Grassland
3) Tropical forest
4) Arctic tundra

24. Which of the following is a physicochemical component of ecosystem?
1) Parasites
2) Predators
3) Water

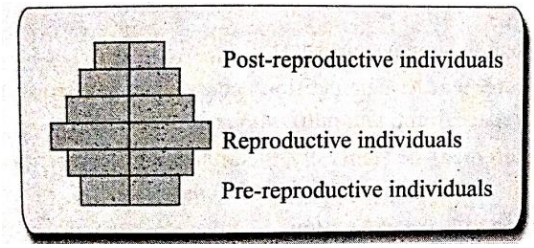
4) Producers

25. Which algae could be found in the deepest water of any water body?
1) Green algae
2) Red algae
3) Brown algae
4) All of the above

Populations

26. Given below are two statements.
Statement I: Under a particular selection pressure organisms evolve towards the most efficient reproduction strategy.
Statement II: Most of the birds and mammals breed only once in their lifetime.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

27. An age pyramid for human population is depicted here.



Which of the following growth patterns is shown by the given age pyramid?
1) Expanding
2) Stable
3) Declining
4) Unstable

28. _____ breeds only once in its lifetime.
1) Tomato
2) Eucalyptus
3) Bamboo
4) Mango

29. Identify the following statements as true (T) or false (F).
A. Tiger census in our national parks is often based on pug marks and fecal pellets.
B. Size of population for any species is not a static parameter.
C. Per cent cover or biomass can never be a meaningful measure to know population density.
D. Population density is necessarily measured in numbers only.
1) A-T: B-T; C-F; D-F
2) A-F; B-F; C-T; D-T
3) A-T; B-F; C-T; D-F
4) A-F; B-T; C-T; D-T

30. Given below are two statements:
Statement I: Immigration is the number of individuals of same species who left the habitat and gone elsewhere during the time period under consideration.

Statement II: Natality and emigration countribute to an increase in population density.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

31. Fill in the blanks.

I. Natural selection operates to evolve the desired traits at the level of (a)_

II. Population ecology links ecology to _(b) and (c)_

1) a-Organism, b-- Habitat, c – Niche
2) a -Community, b - Habitat, c - Population genetics
3) a-Population, b-Population genetics, c- Evolution
4) a -Species, b - Evolution, c - Niche

32. What will be the number of organisms in the population of protozoan which reproduces by binary fission, after six generations?

1) 24
2) 64
3) 32
4) 128

33. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

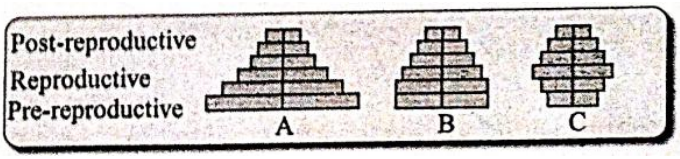
Assertion (A): Evolution of life history traits in different species is currently an important area of research being conducted by ecologists.

Reason (R): Ecologists suggest that life history traits of organisms have evolved in relation to the constraints imposed by the abiotic and biotic components of the habitat in which they live.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

34. Carefully study the given age pyramids, and identify the incorrect statement.



1) Pyramid A depicts rapid increase in the size of a population with time
2) Pyramid B shows negative growth
3) Pyramid B is representing stable population
4) Pyramid C is a graphical representation of a declining population

35. Read the following statements.

A. Exponential growth produces J-shaped population growth curve.

B. Logistic growth occurs when resources are unlimited.

C. For exponential growth, the equation is $N_t = N_0 e^{rt}$

D. Logistic growth produces a sigmoid population growth curve.

1) Only (A) is correct.
2) (A) and (B) are correct

3) (A), (C), and (D) are correct.
4) (A), (B), and (C) are correct.

36. Identify the correct set of statements:

(a) A group of individuals resulting from asexual reproduction is also generally considered a population for the purpose of ecological studies.

(b) Impact of biotic and abiotic factors on population growth can be assessed by studying "intrinsic rate of natural growth.

(c) When resources are not limiting, the growth plot is logistic.

(d) Populations evolve to maximise their reproductive fitness, also called Darwinian fitness (high r value), in the habitat in which they live.

(e) All population of all species in nature has its disposal unlimited resources to permit exponential growth.

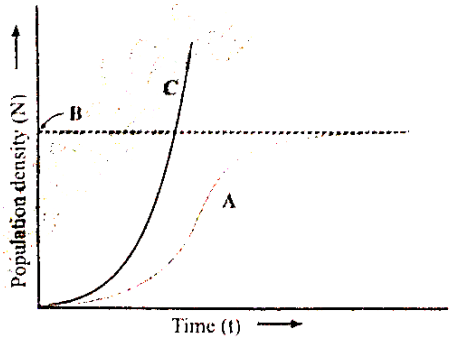
Choose the correct answer from options given below:

1) a, b and d
2) a, b and c
3) b, d and e
4) a, b, c and d

37. The size of the population is represented by its

1) Sex ratio
2) Mortality
3) Natality
4) Density

38. Choose incorrect match for the given population growth curve.



1) $A: \frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
2) B: Carrying capacity
3) C: More realistic curve
4) $C: \frac{dN}{dt} = rN$

39. Lichens are the association between

1) Bacteria and algae
2) Fungus and algae
3) Bacteria and fungus
4) Fungus and virus

40. Interaction between which of the following pairs of organisms is referred to as commensalism?

1) Algae and fungi
2) Lice and human
3) Snake and frog
4) Sea anemone and clownfish

41. Read the given statements.

Statement A: Competition occurs when resources, such as space, light, and nutrient, are in short supply.

Statement B: Generally, the interspecific competition is more intense than intraspecific competition.

Select the correct option.

1) Only statement (A) is correct.
2) Only statement (B) is correct.
3) Both statements (A) and (B) are correct.
4) Both statements (A) and (B) are incorrect.

42. Predators in nature are prudent because

1) The prey species have evolved various defenses to remove the impact of predation
2) If predators overexploit their prey, then prey might become extinct and thus predators will also become extinct because of lack of food
3) In course of evolution, predators have not developed any defensive adaptations
4) Most of the environmental conditions always remain unfavorable to the predators

43. Which of the following interactions between species explains the conduit for energy transfer across trophic levels?

1) Mutualism
2) Competition
3) Predation
4) Commensalism

44. How many of the following shows "+," "-" interaction?

(a) Fig and wasp
(b) Calotropis and herbivores
(c) Sparrow and seeds
(d) Tiger and deer
(e) Orchid on mango tree
(f) Malarial parasite and human

1) Four
2) Five
3) Two
4) Three

45. Select the odd one out wrt mutualism.

1) Lichen
2) Mycorrhizae
3) Cyanobacteria and coralloid root
4) Lice and humans

46. Read the following statements.

A: Some species of insects and frogs are cryptically colored.

B: Some animals are distasteful due to certain chemicals in their bodies.

C: Some animals are poisonous.

The above adaptations are against

1) Predation
2) Parasitism
3) Symbiosis
4) Protection

47. Which of the following means was used by warblers to avoid competition and coexist?

1) Differences in foraging activities
2) Habitat fragmentation
3) Competitive release
4) Both (2) and (3)

48. Select the incorrectly matched pair.
1) Fig and wasp-Mutualism
2) Cuscuta and hedge plant-Commensalism
3) Cuckoo and crow-Brood parasitism
4) Goats and Abingdon tortoise on Galapagos Islands - Competition

49. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The interaction where one species is benefitted and the other is neither benefitted nor harmed, is called commensalism.
Reason (R): An orchid growing as an epiphyte on a mango branch is an example of commensalism.
In the light of the above statements, choose the most appropriate answer from the options given here:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

50. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Herbivores and plants appear to be more adversely affected by competition than carnivores.
Reason (R): Abingdon tortoise in Galapagos Island became extinct within a decade after goats were introduced on the island, apparently due to the greater browsing efficiency of the goat.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

51. Which among the following best exemplifies competitive coexistence?
1) Goats and Abingdon tortoise
2) Balanus and Chthamalus
3) Fourteen species of finches in Galapagos island
4) All of the above

52. Sexual deceit is employed by
1) Ophrys to get pollinated
2) Orchids on mango for shelter
3) Cuckoo and crow
4) Abingdon tortoise in Galapagos island

53. A butterfly of India when resting with folded wings resembles a dead leaf. This mechanism of protection is called
1) Hiding
2) Mimicry
3) Congregation

4) Diapause

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
1	3	2	4	3	4	3	2	1	4	4	4	2	3	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
1	4	3	2	3	1	3	3	3	2	2	3	3	1	1
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	2	4	2	3	1	4	3	2	4	1	2	3	1	4
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**							
1	1	2	1	1	3	1	2							